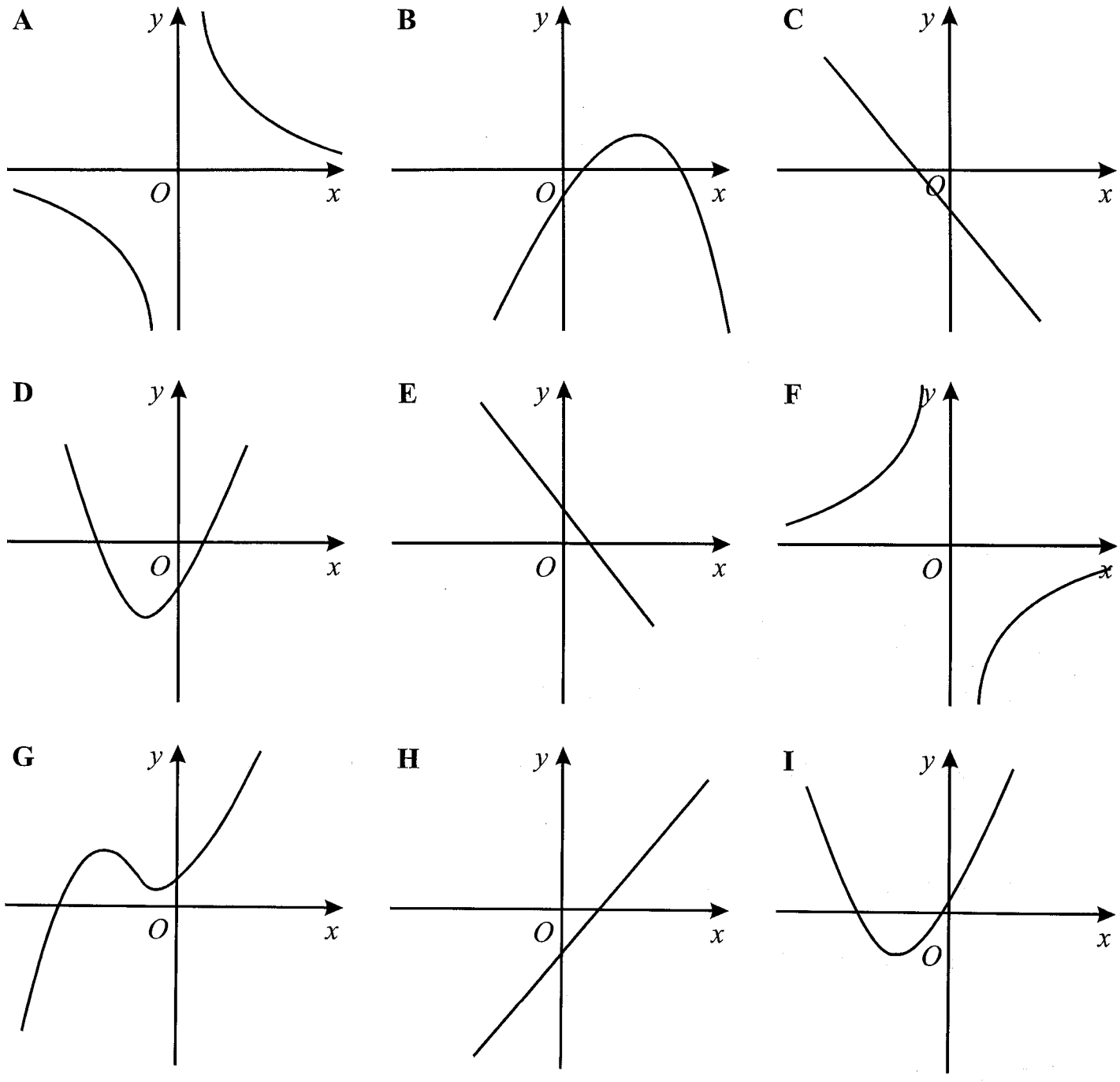
1 Here are nine graphs.



Write down the letter of the graph that could have the equation:

(i) $y = 3x - 2$

H
(1)

(ii) $y = 2x^2 - 5x - 3$

D
(1)

(iii) $y = \frac{3}{x}$

A
(1)

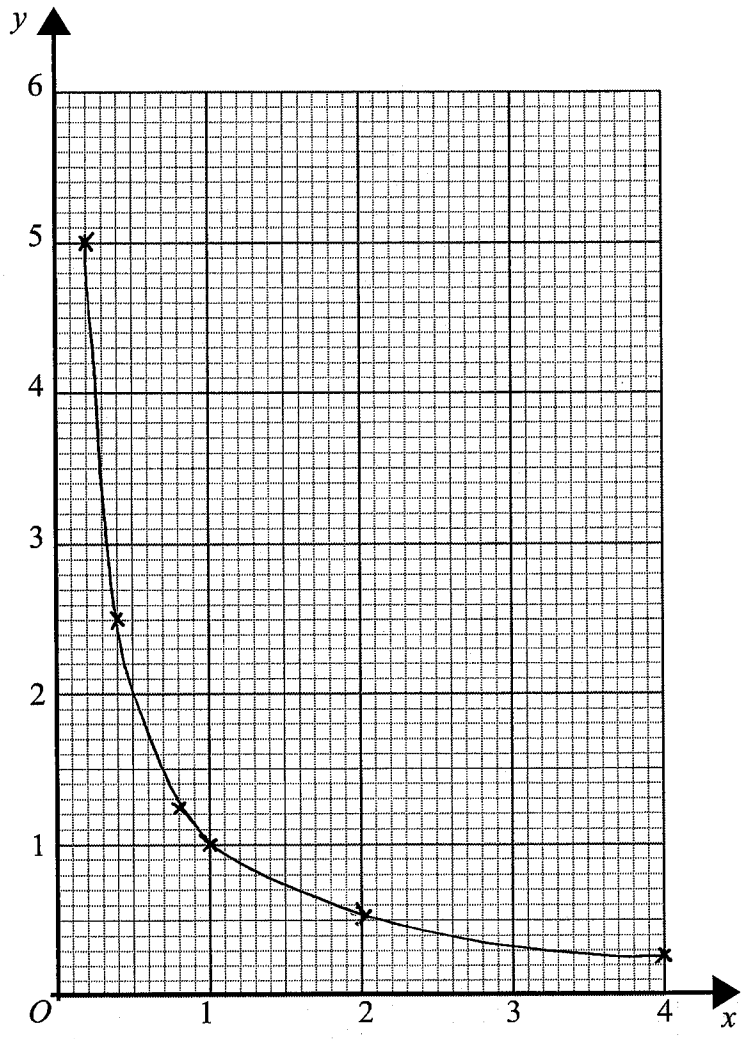
(Total for Question 1 is 3 marks)

2 (a) Complete the table of values for $y = \frac{1}{x}$

x	0.2	0.4	0.8	1	2	4
y	5	2.5	1.25	1	0.5	0.25

(2)

(b) On the grid, draw the graph of $y = \frac{1}{x}$



(2)

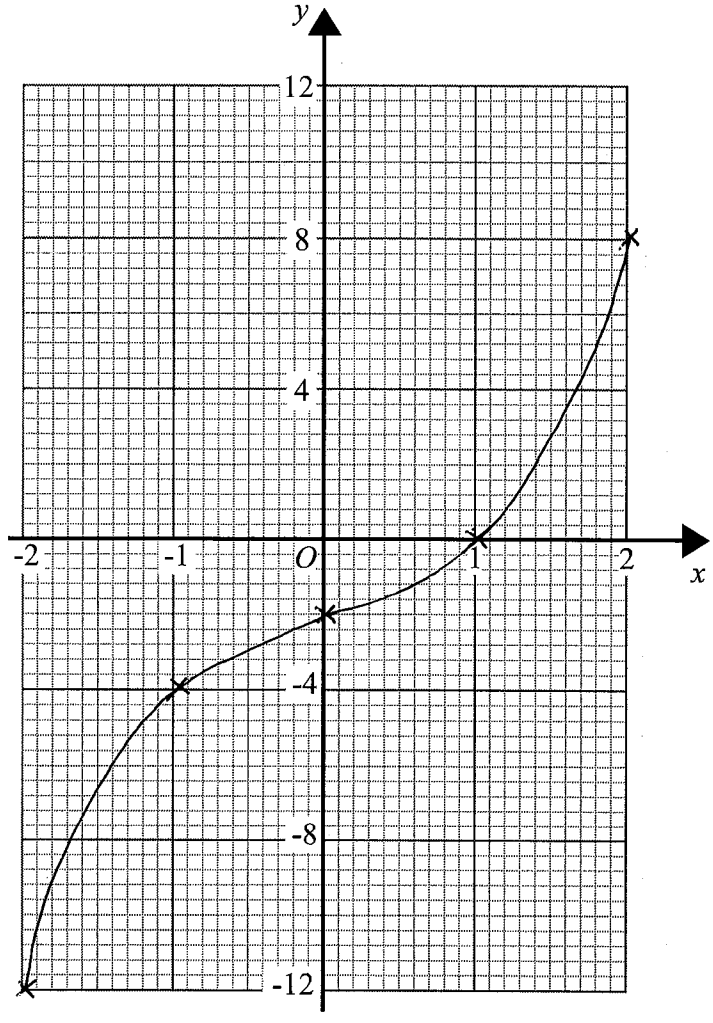
(Total for Question 2 is 4 marks)

3 (a) Complete the table of values for $y = x^3 + x - 2$

x	-2	-1	0	1	2
y	-12	-4	-2	0	8

(2)

(b) On the grid, draw the graph of $y = x^3 + x - 2$



(2)

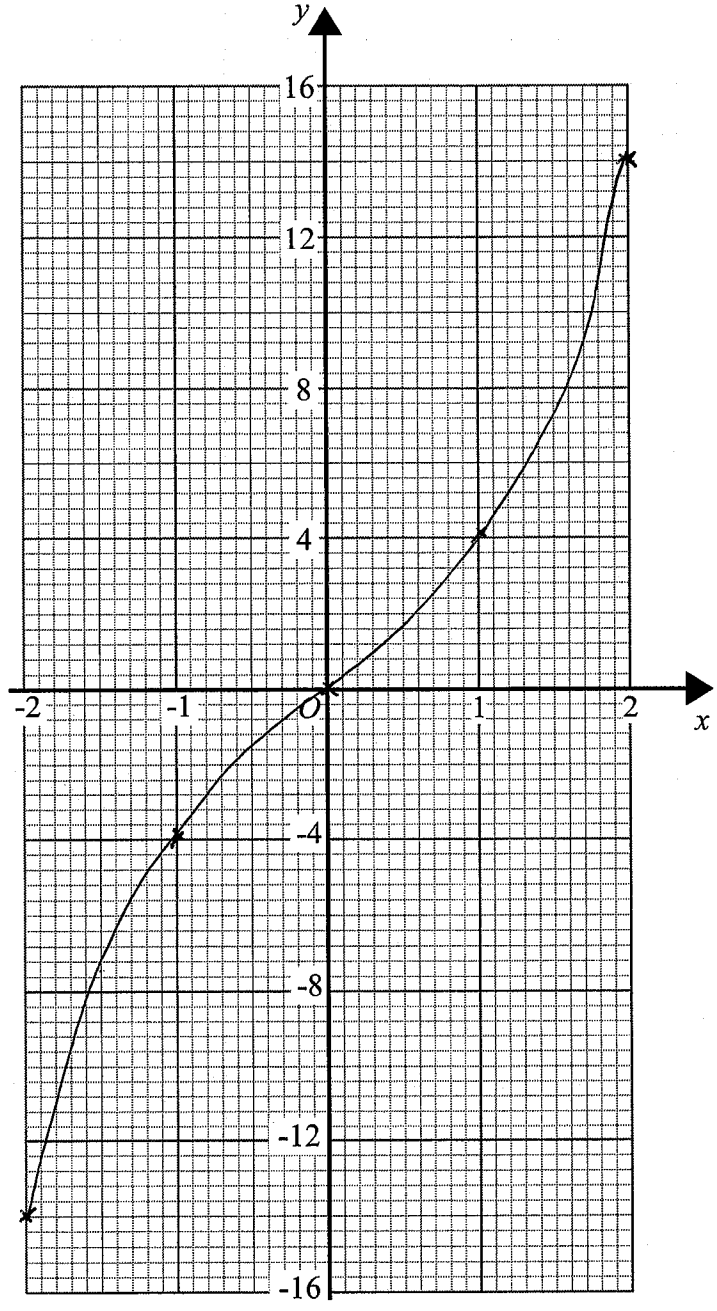
(Total for Question 3 is 4 marks)

4 (a) Complete the table of values for $y = x^3 + 3x$

x	-2	-1	0	1	2
y	-14	-4	0	4	14

(2)

(b) On the grid, draw the graph of $y = x^3 + 3x$



(2)

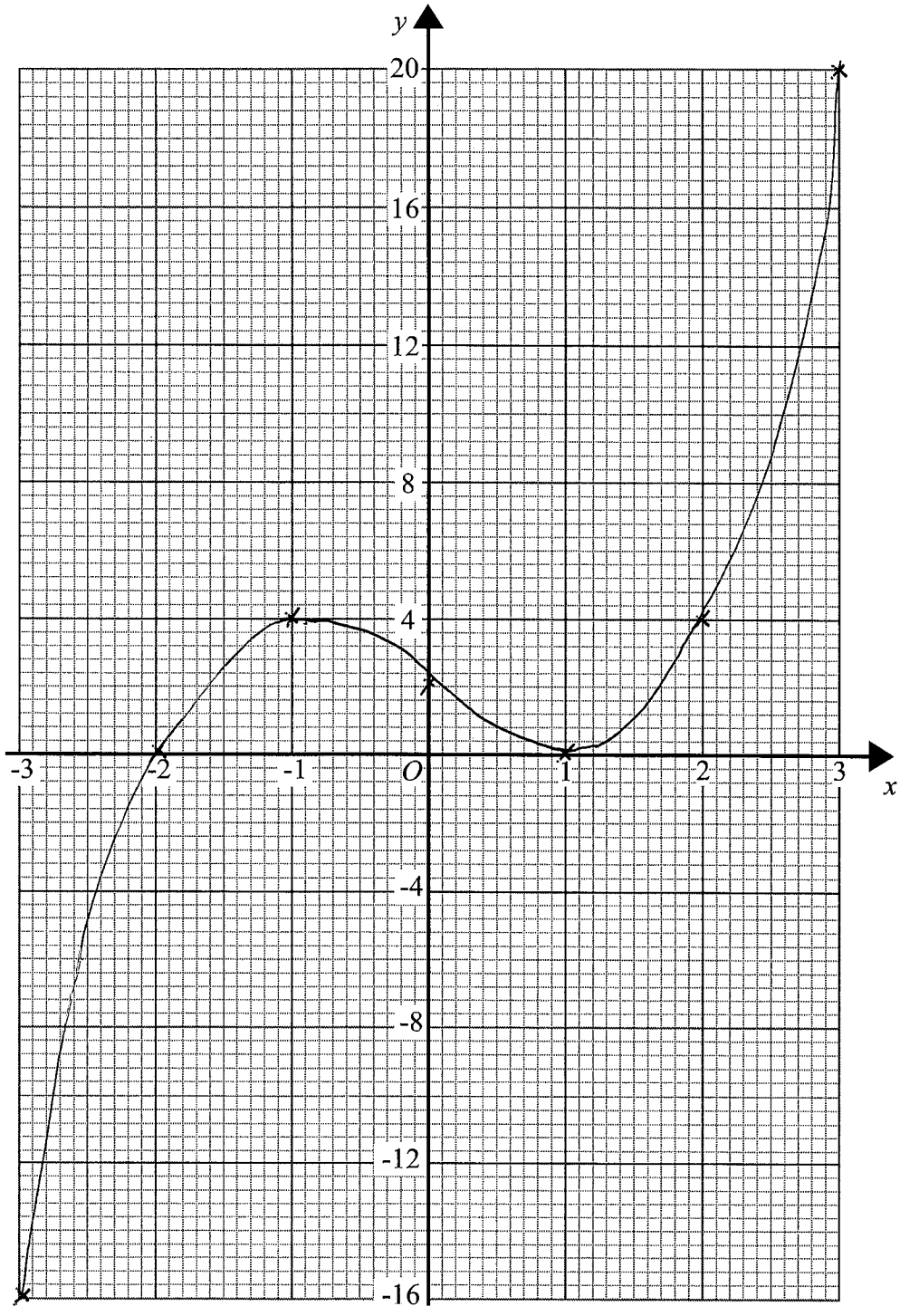
(Total for Question 4 is 4 marks)

5 (a) Complete the table of values for $y = x^3 - 3x + 2$

x	-3	-2	-1	0	1	2	3
y	-16	0	4	2	0	4	20

(2)

(b) On the grid, draw the graph of $y = x^3 - 3x + 2$



(2)

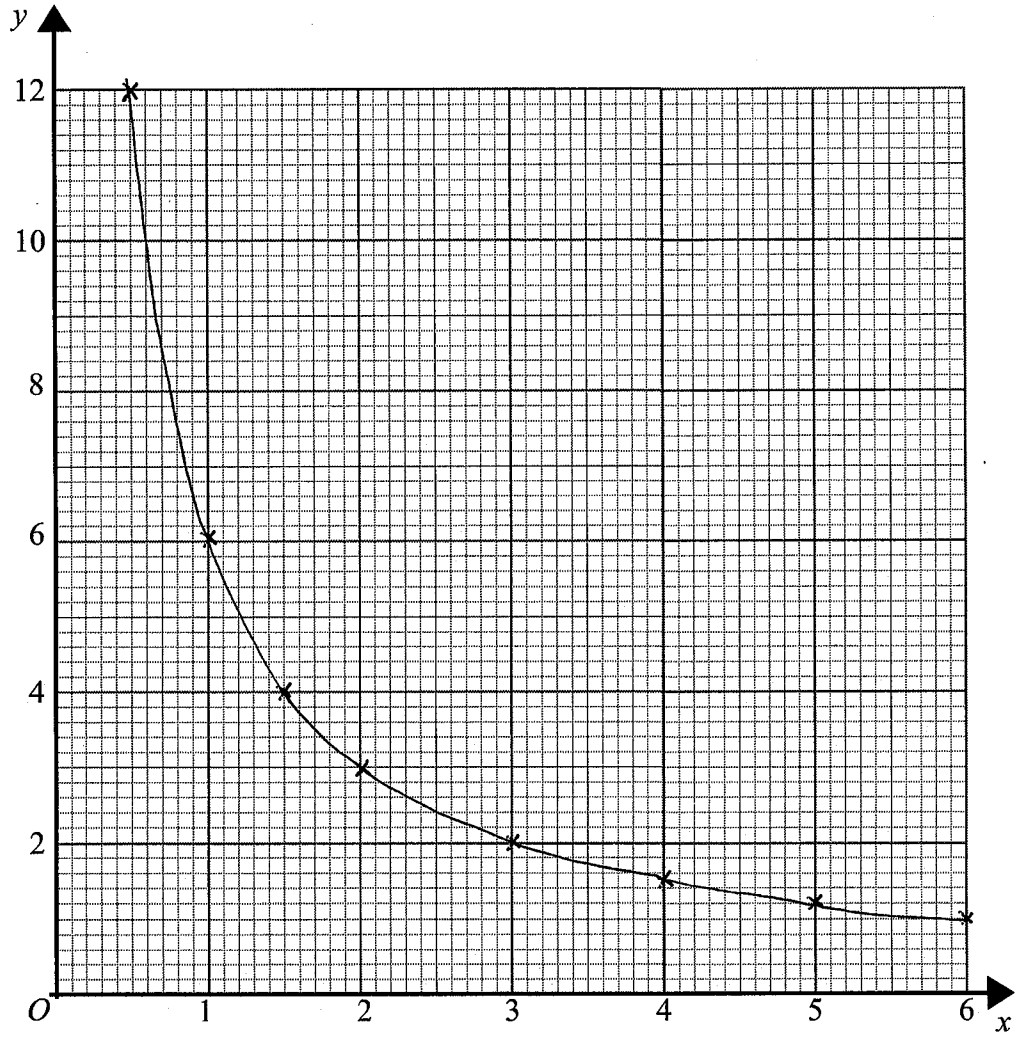
(Total for Question 5 is 4 marks)

6 (a) Complete the table of values for $y = \frac{6}{x}$

x	0.5	1	1.5	2	3	4	5	6
y	12	6	4	3	2	1.5	1.2	1

(2)

(b) On the grid, draw the graph of $y = \frac{6}{x}$



(2)

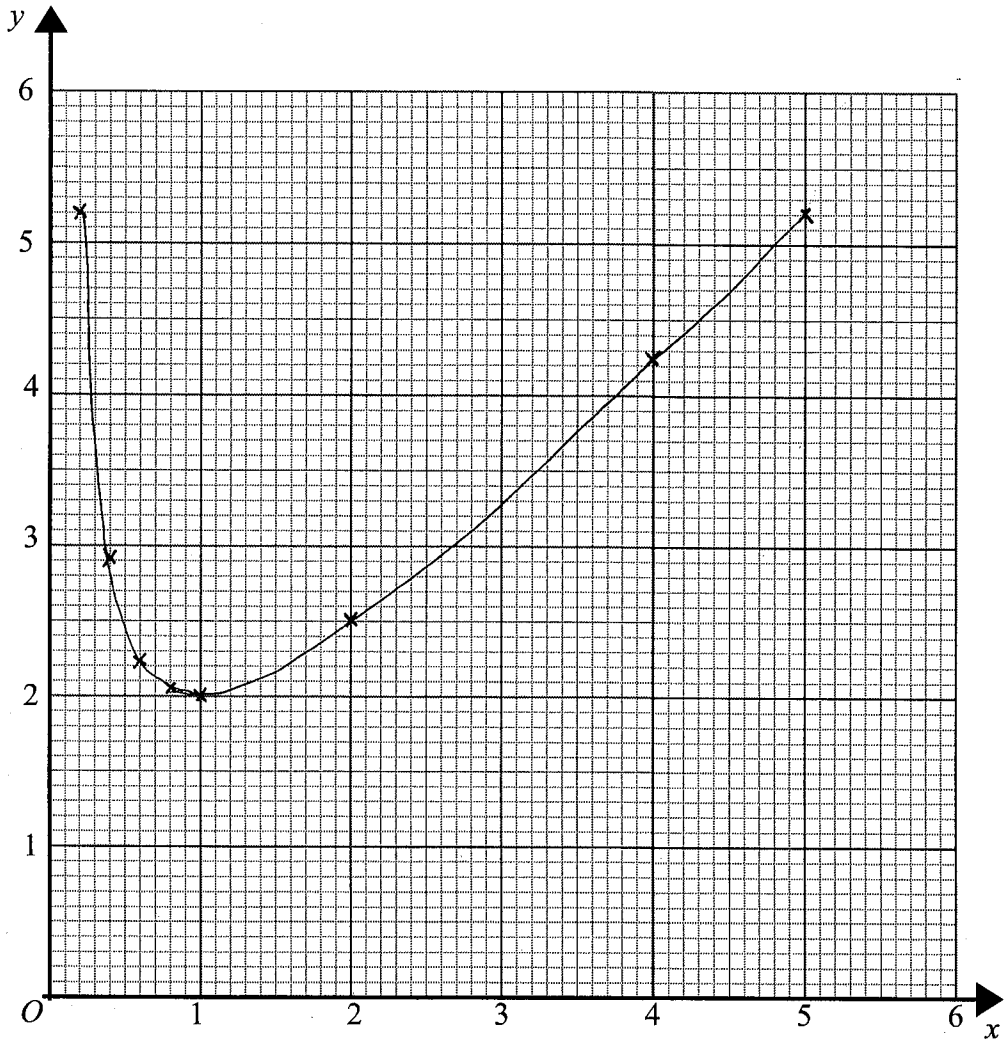
(Total for Question 6 is 4 marks)

7 (a) Complete the table of values for $y = x + \frac{1}{x}$

x	0.2	0.4	0.6	0.8	1	2	4	5
y	5.2	2.9	2.27 (2dp)	2.05	2	2.5	4.25	5.2

(2)

(b) On the grid, draw the graph of $y = x + \frac{1}{x}$



(2)

(Total for Question 7 is 4 marks)